

A Model Cold Climate House

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by John Woodward

The Haida House project came about because people care about housing in a small Indian village. Hydaburg is located on Prince of Wales Island in southeast Alaska, where many of the Haida Nation native peoples live. This is a culturally rich environment; however, the Haida housing situation is not so rich. To address chronic housing problems, Hydaburg Corporation built a local housing program to service their community housing needs.

Doreen Witwer, tribal administrator for Hydaburg, was fed up with the poor quality of life caused by problem housing. She gave Jess Dilts, the village housing director, a green light to spend the time and money that it would take to give the Haidas the skills they needed to make their housing program succeed. After two years of classes in the building sciences provided by the Alaska Building Science Network (ABSN) and myself, the Haidas were ready for a project to showcase their hard-won new skills.

The Haida House project began to take shape when Joe Lstiburek and Betsy Pettit, the principals of Building Science Corporation (BSC), heard me lecture on the lack of good building plans for low-income housing that could be used in places such as Hydaburg, and asked what they could do to help. Because BSC is arguably the premier architecture and engineering firm and building science organization

in America today, I shamelessly took advantage of their generosity. Without their assistance, the revolutionary jump in design demonstrated in this project would not have been possible. BSC contributed the architectural and engineering plans, and provided much of the timely technical expertise needed to transform ordinary building materials into an extraordinary home—the Haida House.

Most houses built in Alaska today are the product of slight modifications to designs imported from milder climates. Such designs seldom perform well in Alaska. Long Alaska winters, high energy costs, and seasonal employment in many of our industries create a demand for building systems engineered from the ground up to provide healthy, functional homes that last a long time while keeping operating costs extremely low. Projects such as the Haida House push the design envelope in every one of these ways.

The most obvious way in which the design of the Haida House differs from conventional construction is the design and placement of the thermal envelope and weather barriers. The Haida House is what might be termed an outside insulation technique (OIT) project. This means that the insulation is installed on the outside of the structural part of the building, not in the structural cavities of the exterior wall or attic space. The floor insulation



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deviates from the OIT technique in that it is placed on the inside of the floor structural decking, rather than on the outside, as would be the case in an all-OIT project. In the Haida House the floor insulation deck was designed for ease of installation and to provide a highly insulated bed for the radiant floor heating system.

From the Ground Up

The Haida House is built on a shallow-draft, pier-type foundation system. The choice of this foundation was based on three factors: stable soil conditions, solid rock close to the surface, and the small project budget. The concrete footers for this system bear directly on bedrock. The foundation consists of three rows of four 6 inch x 6 inch piers, each with a galvanized base and cap.

Site-built horizontal beams bear directly on the vertical piers. The 2 inch x 10 inch floor joists are hung from the center and sidewall beams with galvanized joist hangers on 24-inch centers. This floor-and-foundation system worked fine in Hydaburg, but less-stable soil conditions in many places in Alaska would render such a foundation inappropriate.

One-half inch of oriented strand board (OSB) was installed over the structural deck to support both a

Wall Assembly

The exterior wall framing for the Haida House is 2 inch x 6 inch spruce pine fir (SPF) laminated veneer lumber laid out on 24-inch centers. Even though we did not need cavity space for the insulation layer, we chose the 2 inch x 6 inch option, because it required less labor and fewer materials than 2 inch x 4 inch framing on 16-inch centers.

The house is one and a half stories high. The exterior walls are balloon framed to 12 feet. The one-piece balloon framing for the exterior wall system worked very well and substantially reduced construction time for what is essentially a small two-story building. The second-floor joists are hung at the exterior walls from a ledger board fastened to the exterior wall framing with lag bolts and galvanized framing nails. The wall framing is reinforced at that point with two rows of solid blocking backing up the ledger board. The blocking strengthens the system to accept the added load, and provides the fire blocking.

The thermal envelope is exterior to the wall structural sheathing in this design. The purpose of installing the insulation to the outside of the building is

- to keep the structural framing and sheathing warm and dry over the lifetime of the building, substantially increasing durability;
- to prevent poor thermal performance due to moisture damage;
- to eliminate thermal bridging;
- to provide a continuous insulation layer around the conditioned space;
- to simplify the insulation installation process;
- to enhance insulation installation quality control;
- to move the dew point temperature exterior to the structural components to protect against moisture damage; and



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(top) Handsaws were used to cut the insulation to fit. A power saw would have substantially reduced construction time. (bottom left) An added benefit to the higher density polystyrene is that it tools well. We did not know that when we made plans to router the PEX tubing floor heat distribution system into the floor insulation. (bottom right) The layout for the screw pattern was carefully done to avoid puncturing the PEX heat distribution system laid into the grooves routered into the foam insulation deck. We still managed to puncture one loop in the bathroom, which was quickly repaired.

The Benefits and Challenges of Polystyrene

The permeable and semipermeable properties of polystyrene are desirable for construction in this area. The maritime climate in southeast Alaska means lots of rain with a high wind load. Polystyrene insulation does not lose much performance when wet, and quickly moves moisture out of the system. The expanded polystyrene used in the floor and roof system is high-density (15 psi) open-cell foam. The higher-density foam gives the floor system a better load-bearing capacity than would be provided by the 10 psi, and it makes the floors downstairs feel soft but stable to walk on. Another benefit of the 15 psi polystyrene is that it tools well, unlike the 10 psi polystyrene. The router job went very well with the 15 psi foam.

5 1/2 inch and a 6 inch layer of high-density (15 psi) expanded polystyrene insulation (see “The Benefits and Challenges of Polystyrene”). The load from the walls and roof is transferred through the insulation deck to the structural deck by means of 2 inch x 12 inch floor joists on edge in the

insulation deck. These structural joists are doubled in the rim-band joist areas, and under the bearing wall supporting the upstairs floor. This load transfer strategy provides minimal thermal bridging and creates a clear load path to the foundation beams.

- to layer protection against the high rain and wind load in the northern maritime climate.

The exterior cladding is lap cedar. A drying space behind the cladding was created by using 1 inch x 4 inch vertical stickers attached to the wall structural framing with screws fastened through the insulation layers. These spacer-nailers create a 1/2-inch gap between the cladding and the expanded polystyrene wall insulation. This gap is open to the bottom, allowing air movement to carry moisture out of the system. Cobra mesh installed in the bottom 2 feet of this space discourages critters from moving in.

The wall insulation consists of two layers of 4-inch expanded polystyrene installed in a running bond. The foam used is a 10 psi product of approximately R-4 per inch, yielding a total wall R-value of approximately R-32. No exterior weather barrier was used over the polystyrene. The combination weather barrier, air barrier, and drainage plane is Tyvek Stucco Wrap.

The corner detail on the furring strip-cladding installation took some working out, as the wall exterior insulation depth made hitting corner framing impossible with reasonable fasteners. Dilts used a jig to help router a groove into the foam at the corners to accept horizontal 2 inch x 4 inch nailers. The nailers were screwed through the insulation to the wall framing and installed flush with the foam exterior, to hold the vertical stickers for the cladding installation at the corners.

Window Detail

In the Hyدابurg climate, where winds reach 100 mph, and it can go down below 25°F in the winter, window and door penetrations are detailed to keep the weather out. Head and jamb flashing constructed of bituminous adhesive membrane is installed at all door and window openings. The



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(top and bottom right) A drying space behind the cladding was achieved by using 1 inch x 4 inch vertical stickers attached to the wall structural framing with RS screws fastened through the insulation layers. These spacer/nailers create a 1/2-inch gap between the cladding and the expanded polystyrene wall insulation. This gap is open to the bottom allowing air movement to carry moisture out of the system. Cobra mesh installed in the bottom two feet of this space discourages critters from moving in. (bottom left) The thermal envelope is exterior to the wall structural sheathing in this design, so the wall cavities are within the thermal envelope and available for use to run mechanical systems. This conserves space while not adversely affecting energy efficiency, or durability. Note the stairwell header, and the excess exterior sheathing above the exterior side-wall top plate to accommodate the rafter heel.

Vicor Plus material used to create the bituminous flashing could not be turned onto three planes, so a patch was used to protect the jamb-bottom sill intersection of the rough opening. Flexwrap was not available locally for this project. We hope to use it someday. The patch provided continuous coverage at the bottom corners of the rough opening. A sill dam prevents bulk water penetrating the flashing system from entering the window assembly.

A head flap of the stucco wrap was pulled into place over the bituminous

flashing. The head flap prevents water penetrating the wall system past the insulation layers from intruding into the window opening. The stucco wrap air barrier-drainage plane redirects bulk water outside the system again.

Window jamb extension boxes were site manufactured of yellow cedar. These extension boxes terminate on the surface of the drainage plane, directing water flow back out the graded slope of the sill plate, or to the drainage plane if the sealant used to seal the extensions to the wall fails. The flanges used to install the windows prevented a tight



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the rafters with the screws through the 12 inches of foam. The workers finally used walkie-talkies to communicate between the inside of the home and outside on the roof. Next time we will install solid 2 inch x 6 inch flat blocking between the rafters—creating a much bigger target for the false-rafter fasteners.

We had a problem with insulation drift on the steep pitch. Even though we used mending plates strategically placed around the perimeter of the roofline on top of the roof sheathing and between insulation layers to tie the layers and the sheathing together, we still got some drift. Dilts devised hold-off blocking fastened to the walls to help minimize this drift, but there was still a problem. The gap created at the peak by the slight drift was filled with spray foam. A mismatch at the roof wall intersection was created by the drift of the roof insulation and required trimming. The overall quality of the roof installation was good, but a better plan to prevent insulation drift on this steep pitch is still needed.



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(top) Ready to put the lid on! (bottom left) Head and jamb flashing constructed of bituminous adhesive membrane are installed at all door and window openings. (bottom right) I recommend in future using windows installed without the use of flanges.

seal in this area. (I recommend using windows installed without the use of flanges in the future.)

Roof

The roof system is constructed as follows from the inside out:

1. Site-built structural rafters are installed with tails flush to the wall line.
2. Five-eighths-inch OSB structural sheathing is installed over the rafters.
3. Tyvek Stucco Wrap is installed over the sheathing.

4. Two 6-inch staggered layers of high-density (15 psi) expanded polystyrene (approximately R-4.5 per inch for a total R-value of R-54) are installed over the Tyvek.
5. False rafters (2 inches x 4 inches) are mounted on edge on plywood ribs fastened through the insulation to the structural rafters. These provide overhang and a drying-runoff space.
6. A layer of 1/2-inch exterior plywood installed over the false rafters serves as a base for the tar paper and the metal roofing.

The false rafters were fastened to the structural rafters with 14-inch screws. It was quite a problem hitting

Space Heating

The heating, domestic hot water (DHW), and ventilation systems in the Haida House had several unusual features. The high levels of thermal insulation in the home design made a radiant floor heat distribution system quite effective and relatively foolproof. The goal was to make the systems as simple, inexpensive, efficient, and durable as possible, utilizing radiant floor heat distribution. Phil Kalusa was the technical advisor for the heating and ventilation systems.

The primary heat source and DHW are both provided by a direct-vent Bock model BCS 32-gallon tank-type water heater. It has a standard Beckett oil burner, which is easily maintained. The water heater is a (semi) direct-vented unit that shouldn't backdraft. It has no barometric damper, which saves energy, especially in the windy climate of Hydaburg.

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The efficiency of the Bock water heater as a combo unit is not all that impressive, probably in the 80%–82% AFUE (annual fuel utilization efficiency) range, but given the limited amount of heating required by the very highly insulated Haida House, less energy is lost than would be the case with some of the newer high-tech systems in a less insulated house. The builders decided that simple and reliable trumps a little improvement in efficiency. The \$1,400 cost of the unit allows for future replacement as proven new technologies to service small heat loads become available.

The small Taco 007 circulation pump uses approximately 60 watts. A 120V thermostat installed in the main living area provides temperature control and also controls the circulation pump. Individual room temperature control can be accomplished by tweaking the flows in the individual loops at the manifolds, though given the high thermal performance of the building shell, indoor temperatures will probably remain pretty steady.

Cross-linked polyethylene (PEX) tubing was used for piping for the radiant system. It was not possible to install the PEX tubing using a staple-up approach on the main floor, because the insulation deck was inside the structural deck in this design. So before the plywood subfloor was installed, grooves were routed into the foam to hold the PEX. Finding a good layout for the routed grooves took a while. The challenge was to get two runs inside of each joist bay while allowing room to fasten the insulation decking down to the structural deck joists without puncturing the PEX. We found the PEX could make an 8-inch bend. Anything smaller caused it to kink; anything larger and we could not get two runs per joist bay. We made a plywood jig to make the radius at each run end. Eliminating the baseboard, which tends to get beat up in low-income housing, made the radiant floor a good option and saved some valuable floor area.



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(top) The roof system is on a 12/12 pitch. Here the drainage plane is complete—temporarily. (bottom left) The workers had to use walkie-talkies to communicate between the inside of the home and outside on the roof. Next time we will install solid 2 inch x 6 inch blocking between the rafters—creating a much bigger target for the fasteners. (bottom right) The roof was the best architectural feature of the home, because of the great amount of space created for a story and a half.

Supply and return manifolds were installed inside the washer-dryer closet space, to serve four circulation loops on the main floor and three loops on the second floor. Flow through the loops is controlled by manually operated valves located on the supply and return manifolds. The manifolds are wall mounted in the downstairs utility room. There are also purge and fill valves installed on these manifolds.

For the sake of simplicity, potable water is directly circulated through the radiant floor tubing. All components are potable water compatible. For added safety, a timer is wired into the circulation pump so that the pump runs for a few minutes every day to purge the water. Hot water is drawn from the tank through a side port

on the tank and the return water is plumbed into the bottom drain line port.

After several hours of operation, the floor temperatures ranged from 71°F to 74°F, while the ceiling temperatures ranged from 66°F to 68°F. Outdoor temperatures were in the mid- to high 40s. All indications are that the system is working as planned.

Domestic Hot Water and Plumbing

The Bock is used to provide the DHW. The water enters the home through the floor in the utility room. From there it goes to the plastic Vanguard domestic water distribution manifold, which distributes cold water



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The wind was a factor throughout the project. Sustained wind-speeds of approximately 80 mph plus happened on at least three occasions during construction. One storm brought gusts to 103 mph.

Table. Heat Recovery Ventilator (HRV) Air Flow

	Exhaust (CFM)	Supply (CFM)
Max.	93.5	87
High	79	79
Med.	61	61
Low	35	50

to the lavatory, sinks, and shower. The water to be heated passes through the cold water manifold, goes on to the Bock, is heated, and then returns to the hot water distribution manifold. All domestic water lines are one-piece PEX routed through manifolds, which should prevent pressure drops when several fixtures are being used at once.

Ventilation

A small heat recovery ventilator (HRV) (Fantech model SH704) controls the ventilation in the Haida House. This ventilator is small and quiet, and could easily be retrofitted with a more-efficient fan and smart controls. The unit was selected based upon its net air and power consumption at low speed. Sensible recovery efficiency is a bit lower than that for other models at approximately 57% on low speed. The high-quality

blowers provide low flows at minimal energy use and are very quiet. The unit comes without any speed controls but easily accepts a standard speed controller.

A simple mechanical boost timer was installed in the bathroom to override the low-speed controller. Defrost is accomplished by shutting off the supply air fan for a timed period when outdoor air temperatures fall below 25°F. This defrost strategy eliminates the need for a defrost motor and damper assembly,

which would require considerable maintenance.

The unit was installed on the second floor in the mechanical room. This space provided a good central location for both the interior ductwork and the insulated ductwork to the exterior. A nearby drainpipe provided easy access for the condensate drain. The exhaust air from the HRV is routed to the outside directly out the gable end of the wall, which is on the lee side of the prevailing winds. The supply air intake was routed down through the main floor. A 14 inch x 20 inch high-efficiency pleated furnace filter is boxed in between the open floor joists above the supply duct to filter the outdoor air. This will greatly reduce the need to clean the HRV internal filters and core.

The location and standard size of the filter box make it easy to replace the filter when necessary. With the large filter area and small flows, replacement should not be necessary for five to ten years. Flow restriction will be minimal. Also, the location under the home, which is unconditioned and wide open, avoids a wall penetration and is fairly wind pressure neutral, which is critical in such windy locations.

Exhaust ductwork was provided to the bathroom and kitchen. Supply ductwork was provided to each of the three bedrooms. Fresh air to the living room was not provided, because it was not cost-effective to route ductwork to the living room. The open layout of the home provides for adequate circulation.

Power consumption on the HRV on high was 39 watts. We were unable to measure power consumption on the lower speeds. Previous testing on other similar models was in the 18-watt range for low speed.

Balancing stations were temporarily installed and flows were checked (see Table 1). Flow dampers were deemed unnecessary. At medium speed, the speed at which the unit is normally run, the HRV showed a balanced flow (at least for that day, under those particular wind conditions). Though no grilles had yet been installed, flow was detected in all supply and exhaust ports.

A depressurization test on the range hood found an 8–10 Pa depressurization when the range hood was on high. Turning the HRV on and off during the test had very little effect on depressurization.

Is it Affordable?

Total cost for the home, materials and labor, was \$127,000. Projected fuel use for space heat is 68 gallons per year, costing approximately \$200 with current prices. Estimated fuel use for DHW is 187 gallons, or about \$550 for the year. Total construction cost for the home is approximately \$90/ft². Akwarm Home Energy Modeling Software predicts that adjusting the depth and type of the foam insulation within reasonable limits will make it possible to reach the same energy target anywhere in the state. The flashing and airtightness details are consistent with use in many parts of the United States.

Monitoring

A dryer vent was installed in October 2006. The penetration of the building during installation provided

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an opportunity for us to measure the moisture content of the subfloor and soffit OSB, and visually inspect for moisture between the layers of floor foam. No moisture was observed. The moisture content of the subfloor was 12% on both sides, and the moisture level of the exterior exposed soffit OSB was 15%. The exposed floor joists were also in the 15% range. Inspection of other previous penetrations through the floor found no signs of moisture, indicating that the floor system has dried from the construction wetting period.

Next House

As with any first-time project, we learned a lot that will help us the next time.

- The steep pitch of the roof is both the home's best asset and the hardest architectural feature to build. A system to build the roof on the ground and hoist it into position, in sections if necessary, would make the roof construction safer, and building it on the ground would cut labor costs significantly.
- Utilizing a Triodetic Space-Frame for the foundation system would save time and provide in-plane support in poor soils.
- Creating the floor insulation deck as a site-manufactured structural panel, and changing the finish floor support system to an open-web truss inside the insulation deck, would provide room for utilities and keep the structural floor system warm and dry over the life of the building.
- Free-spanning the second-floor joists with an open-web truss would eliminate the structural joists in the field in the first-floor insulation deck, and would also make it possible to use the open web space for ductwork, water pipes, and electrical conduits. This system would use half as many hangers as the present design, and it would save time on installation.



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This photo was taken on a nice day for roofing. The metal went on as expected; it was pretty straightforward.

- Routing the PEX floor heat distribution into the floor foam was labor intensive. A simpler heat distribution system could be utilized to save labor costs and reduce the possibility of penetrations. Using a freestanding space heater or combining ventilation and heat distribution are possibilities.
- In the more remote northern regions, care must be taken to provide independent backup heat as a redundant safety measure. Also, renewable-power systems, such as wind generators and PV, integrally installed as new construction, are necessary where wind and solar regimes permit, to offset operating expenses, which currently run \$4–\$8 per gallon for heating fuel, and 55¢–58¢/kWh in many communities off the road system in Alaska.

This time, we did the best we could with the project budget we had. The result is a very energy-efficient well-built house with an extremely robust envelope and a comfortable floor plan drawn to utilize the living space very effectively.

John Woodward is the owner of Alaska Building Performance Specialists and was the project consultant for the Haida House.

I offer many thanks to Phil Kalusa of Arctic Energy Systems, who provided the design for the heating and ventilation systems, for his technical expertise in training the Haidas to do the installations. I paraphrased or quoted verbatim from his mechanical reports in creating this article. Marquam George, from the University of Alaska Southeast Juneau, put some time in as well, helping out with the blower door testing and installing the monitoring equipment. Marquam is a pioneer in creating buildings of this type in Alaska and can always be counted on to do excellent work.

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